

Quiz 2 Class 2 Solutions

Computer Security Fundamentals Spring 2026

Instructions: Answer the questions, **closed-book, closed-notes**. Time: 10 minutes

1. (**T**) By itself, symmetric encryption cannot solve the problem of sharing a secret key in the presence of an eavesdropper. (True/False, 2 pts)

2. (**C**) Bob sends Alice a signed message using asymmetric encryption. How should Alice verify the signature? (2 pts)

- A. Verify the signature with her public key
- B. Verify the signature with her secret key
- C. Verify the signature with Bob's public key
- D. Verify the signature with Bob's secret key

3. (**D**) Almost all computers and smartphones are preconfigured with (2 pts)

- A. The device's public and secret key pairs
- B. The device's public key
- C. The root CAs' public key and secret key pairs
- D. The root CAs' public key

4. (Short Answer/Calculation, 4 pts) Your classmate posted the hash of:

(his credit card number (16 digits) $\oplus$ PIN number (4 digits)) on TikTok/Douyin and said "This is the SHA-256 hash of my credit card number and PIN number combination. You cannot know the number because hash is one-way." Is this true for the following two conditions?

Assume calculating SHA-256 hash of a 20-digit number on an average PC takes 1 microsecond (10^{-6} s), and numbers 0-9 for every digit are equally likely.

1) The attacker doesn't know anything about his credit card.

2) The attacker knows his credit card number.

Note: The $\oplus$ symbol means concatenation, $12345678 \oplus 910 = 12345678910$

Solution: For both conditions, the attacker can try something like a dictionary attack, calculating the hash of all possible digit combinations until he sees a match.

- 1) 10^{20} total combinations for the 20-digit number, brute forcing all the possible combinations takes 10^{14} seconds. You should know that 10^5 seconds are roughly a day, so it takes 10^9 days or at least a few million years to try all combinations, so the classmate is technically true.
- 2) In this case, all 16 digits in front are known, and the attacker just need to try 10^4 combinations. Brute forcing takes only 10^{-2} seconds or 0.01s. The classmate is definitely wrong.